INSTITUTE OF INFORMATION TECHNOLOGY & MANAGEMENT, NEW DELHI

Date: Jan 02, 2026

Major Project Guidelines for BCA-VI semester

Objective

All the students enrolled for BCA Programme, are to carry out a software project as part of curriculum. At the end of the semester the students are to submit a written project report to be forwarded to GGSIP University. The aim of the project is to give the students an integrated experience in solving a real-life problem by applying knowledge and skills gained on completion of theory papers and in-house practical papers during BCA course. It provides an occasion for students to realize the importance of resource and time management, ownership of task towards deliverables, innovation and efficiency in the task management. It also provides a good opportunity for students to build, enhance and sustain high levels of professional conduct and performance and evolves a problem solver frame of mind in students at early stage. It also prepares students for taking up responsible assignments in the corporate establishment.

Programme Objective:

Program Objectives (PO's)	Description
PO1	Understand the fundamental concepts of Computers, Software hardware and Peripheral devices and evolution of computer technologies.
PO2	Familiarized with Business environment and Information Technology and its Applications in different domains.
PO3	Gain knowledge to identify, explain and apply functional programming and object-oriented programming techniques and use of databases to develop Computer programs.
PO4	Analyze, design, implement and evaluate computerized solutions to real life problems, using appropriate computing methods including web applications.
PO5	Understand the front end and backend of software applications.
PO6	Gain expertise in at least one emerging technology.
PO7	Acquire knowledge about computer networks, network devices and their configuration protocols, security concepts at various level etc.
PO8	Apply techniques of software validation and reliability analysis to the Development of computer programs.
PO9	Acquire Technical, Communication and management Skills to convey or present information, applications, instructions, policies, procedures, decisions, Documentations etc. verbally as well as in writing.
PO10	Recognize the various issues related to society, environment, health and vivid cultures and understand the responsibilities to contribute in providing the Solutions.
PO11	Acquire technical skills to lead a productive life in the society as a professional or as an entrepreneur.

Programme Learning Outcome:

PLO	Attribute	Description
PLO1	Communication Skills	Understand the fundamental concepts of Computers, Software Hardware and peripheral devices and evolution of computer technologies.
PLO2	Use of Software Tools	Familiarized with Business environment and Information Technology and its Applications in different domains.
PLO3	Technical Skills	Gain knowledge to identify, explain and apply functional programming and object-oriented programming techniques and Use of databases to develop computer programs.
PLO4	Domain Awareness	Analyze, design, implement and evaluate computerized solutions to real life problems, using appropriate computing methods including web applications.
PLO5	Technical Support	Understand the front end and backend of software applications.
PLO6	Analysis and investigation of Complex Computing Problems	Gain expertise in at least one emerging technology.
PLO7	Design / Development of Solutions	Acquire knowledge about computer networks, network devices and their configuration protocols, security concepts at various level etc.
PLO8	Imbibe Cyber Ethics	Apply techniques of software validation and reliability analysis to the development of computer programs.

Course Outcomes	
CO#	**Description**
CO1	Apply Knowledge, cognitive & problem solving skills to real life problems
CO2	Demonstrate leadership skills with team and individual work with innovation & efficiency in task management
CO3	Effective communication & Presentation skills with efficient articulation of information
CO4	Demonstrate productive and social skills to be employed in corporate establishments

University Scheme for Summer Training & Project

As per the syllabi of BCA (Paper Code: BCA-308), students of Semester VI are to write a project report comprising of 06 credits. The project report has two components, viz.

(a) External : Project (60 Marks), where a written report is to be submitted. It involves external viva and presentation.

(b) Internal : Project (40 Marks), it includes continuous evaluation from idea generation to final presentation.

Scope of Project

It is partly the responsibility of the student to find a relevant topic for his/her project and decide it in consultation with the guide allocated to him/her. The project work shall be an application development in a language/platform that the student has learnt during the previous semesters. It shall be an in-house development. **Each student is required to carry out the independent work and submit the report individually.**

NOTE: ANY PREVIOUS WORK OR BORROWED REPORT WILL BE SUMMARILY REJECTED AND IN ALL CASES OF REJECTION THE WORK IS TO BE REPEATED AFRESH. STUDENT MUST ENCLOSE PLAGIARISM REPORT OF THEIR PROJECT TO AVOID DUPLICACY OF WORK.

Methodology

General Guidelines that are to be performed to conduct the Major project are given in details in **Appendix-A**.

Format & Final Report

The student will submit his/her project report/thesis in the prescribed format, as given hereunder, after the second presentation. The project report/thesis should include:

a) ONE hard copy (Maroon color with golden print) of the project report / thesis.
b) Soft copy of project including all resource code/ compiled binary code and the manuscript in MS-Word document format.

The Guidelines for the format to be used for compilation of the project report are given in Appendix**-B**. All students are to adhere to these guidelines.

Appendices

The appendices are to be attached at the end of the report and to be numbered as Appendix-A, Appendix-B etc right justified at the top of the page. An appendix may have annexure (s). If there are annexure, these are to be attached immediately after the said appendix. The annexure are to be numbered as Annexure-I, Annexure-II, Annexure-III and Annexure-IV etc.

Schedule of Submissions

Students are required to strictly follow the schedule given below:

To be Completed by Date	Activity	Marks Allotted	Remarks
Jan 08, 2026	Briefing about the Major Project	-	-
Jan 16,2026	Finalization of Topic		
Jan 27, 2026	Phase I: Project Synopsis / Proposal Evaluation(CO1,CO2)	10	As per Appendix-A.
Feb 20,2026	Phase II: Completion & Submission of Activities under Chapter 1 & 2(CO1,CO2)	10	As per Appendix-A.
Mar 14 ,2026	Phase III: Completion & Submission of Activities under Chapter 3 & 4(CO1,CO2)	10	As per Appendix-A.
Mar 28 ,2026	Phase IV: Submission of Final Project Report(CO3,CO4)	10	As per Appendix-A.
Apr 11,2026	Phase V: Project Presentation & Viva-Voce(CO3,CO4)	40	As per Appendix-A & B.
	External Presentation & Viva	**60**	
Total Marks		**100**	

Ms. Leena Gupta
Project Coordinator

Dr. Ganesh Kr. Wadhwani
HOD-CS

Prof (Dr.) Rachita Rana
Director

Copy to:
1. All Students of BCA VI (M+E)
2. All Guides

Appendix 'A'

GENERAL GUIDELINES FOR THE PROJECT WORK

1. The project should be original, of real-life value, and not copied from existing material from any source. A student should ensure that he understands what is expected by preparing a requirement document of his understanding and get it reviewed by the guide.
2. Design document should also be reviewed and code should also be peer reviewed.
3. A user manual has to be prepared and reviewed.
4. Testing has to be thorough and at various levels, followed by an acceptance test based on the requirement document and user manual.
5. Students should follow the steps as discussed in Software Development Life Cycle while writing dissertation and use Software Engineering Methodologies for development of deliverables, mere programming will not be sufficient. Students must note that interviewers for job are often more interested in the problem solved, alternatives that could have been tried and the benefits derived from the developed application, rather than just implementation details.
6. One Project will be submitted only by one student. However, if the nature of the project is very big & large enough to be divided in different independent big modules having an estimated required effort of minimum 06 weeks to be developed by one person, can be taken up and designed in such a way that every student will be responsible for one module and will submit only that specific module as it were a complete software project.
7. Every student has to get his / her synopsis approved from the guide.
8. The synopsis must be brief i.e., not more that 4-6 pages. It must address details like (however, students may follow the SRS format of IEEE for writing Synopsis)
    a. Name / title of the project,
    b. Statement about the problem,
    c. Why the Particular topic is chosen? It must address Present State of the Art.
    d. Objective and scope of the project,
    e. Analysis, Design, Development & Testing Methodology,
    f. H/W& S/W to be used,
    g. Testing Technologies to be used,
    h. What contribution / value addition would the project make?
    i. Limitations / constraints of the project,
    j. Conclusion, Future Scope for Modification,
    k. References and Bibliography.
9. After approval of the Synopsis, Students will need to give Two Presentations / Demonstration, as per the schedule fixed by their respective Institutions or University. First Presentation will ideally be given after Design Phase is over. Second should be given when System Testing is over. First Presentation should be given in the first month. Second Presentation (Projects) is in second month. Final Project reports should be given latest by the end of the third month of the semester.

Appendix-B

FORMAT OF THE PROJECT REPORT

Format

Note the following guidelines with respect to Preparation of the Documentation. Please note that documentation is meant for other people, and hence it must be self-explanatory, in all respect. The sequence of the Pages in the Project Report will be as follows:

FORMAT OF THE STUDENT PROJECT REPORT ON COMPLETION OF THE PROJECT

Synopsis

1. Title of Project
2. Problems with the Existing System(Note: Consider the previously existing project as Manual)
3. Description of the Proposed System(Note: The proposed (new project that you are making) system will be web based, online system)
4. Description and identification of the Functional Modules(Here, mention the name of the modules and the users of the system)
5. Tools/Platforms
   5.1. Hardware specification (RAM, HDD, Processor etc.)
   5.2. Software specifications (OS, Front End, Back End, Any other additional s/w Interface reqd.)
6. Methodology
   6.1. SDLC Model to be used
   6.2. Justification for the Selection of Model
7. Future Scope

FORMAT OFPROJECT REPORT

I. Cover Page as per format
II. Acknowledgement
III. Certificate of the project guide
IV. Synopsis of the Project
V. Main Report
   i. Objective & Scope of the Project
   ii. Theoretical Background Definition of Problem
   iii. System Analysis & Design vis-a-vis User Requirements
   iv. System Planning (PERT Chart)
   v. Methodology adopted, System Implementation & Details of Hardware & Software used System Maintenance & Evaluation
   vi. Detailed Life Cycle of the Project
      a. ERD, DFD
      b. Input and Output Screen Design
      c. Process involved
      d. Methodology used testing
      e. Test Report, Printout of the Report & Code Sheet

VI. Coding and Screenshots of the project
VII. Conclusion and Future Scope
VIII. References

Title Page

The format of the title page is attached as **Annexure-I.**

Certificate

The format of the certificate is attached as **Annexure-II**. A certificate by the student, guide and duly authenticated by the Directors to be attached.

Acknowledgement

In the "Acknowledgement" page, the writer recognizes his indebtedness for guidance and assistance by the guide and other members of the faculty. Courtesy demands that he/she also recognizes specific contributions by other persons or institutions such as libraries and research foundations. Acknowledgements should be expressed simply, tastefully, and tactfully duly signed above the name.

TEMPLATE FOR CHAPTER SCHEME
The format of 'Chapter Scheme is attached as **Annexure-III**.

FORMAT OF LIST OF TABLES AND FIGURES

The format of 'list of Tables/Figures/Abbreviation and References/Bibliography is attached as **Annexure- IV.**
BODY OF THE PROJECT REPORT: GUIDELINES FOR PROJECT REPORT WRITING
The pages coming under the preview of the CONTENTS will only be numbered in the BOTTOM of the Page Centrally Aligned.

1. ONE Hardbound Copies (One Original and Two Xerox) will be submitted with the Institute out of which one will be given back to the candidate. All the students are required to follow the same binding format in maroon color with back quote mentioning title of the project, name of the student and year.
2. At the end of the Project Report Two White blank sheets must be attached.
3. At the beginning of each chapter one blank page (Strictly as per the sample supplied) must be attached. These pages will neither be numbered nor counted in total numbering of pages. They will only indicate the beginning of a New Chapter with its learning objectives.
4. Font size of the documentation will be 12 Times New Roman and the pages will be one and half line spaced.
5. Good quality white A4 size paper should be used for typing and duplication. Care should be taken to avoid smudging while duplicating the copies.
6. Page Specification: (Written paper and source code)
    - Left margin - 3.0 cms
    - Right margin- 2.0 cms
    - Top margin 2.54 cms
    - Bottom margin 2.54 cms

- Page numbers - All text pages as well as Program source code listing should be numbered at the bottom center of the pages.

7. **Normal Body Text: Font Size**: 12, Times New Roman, Double Spacing, Justified. 6 point above and below Para spacing
8. **Paragraph Heading Font Size:** 14, Times New Roman, Underlined, Left Aligned. 12 point above & below spacing.
9. **Chapter Heading Font Size:** 20, Times New Roman, Centre Aligned, 30 point above and below spacing. **Coding Font size :** 10, Courier New, Normal
10. **Submission of Project Report to the University:** The student will submit his/her project report in the prescribed format. The Project Report should include:
    1. One copy of the summary/abstract.
    2. One hard Copy of the Project Report.
    3. The Project Report may be about 75 pages (excluding coding).
11. Kindly note that all-methodological details and theoretical aspects must be written in students‟ own words. Copying from books or other students will not be accepted, in any case.

    *******All students are informed not to write definition of various concepts in index, mention the topics w. r. t. to your project (i.e., how various concepts have been implemented in the project) * No detailed theories required.*******

INSTITUTE OF INFORMATION TECHNOLOGY & MANAGEMENT
Accredited 'A' Grade by NAAC & Recognised U/s 2(f) of UGC act
Rated Category `A+' by SFRC & `A' by JAC Govt. of NCT of Delhi
Approved by AICTE & Affiliated to GGS Indraprastha University, New Delhi

Annexure-I

Title of the Project Report
(Times New Roman, Italic, Font size = 24)

Submitted in partial fulfillment of the requirements

for the award of the degree of

Bachelor of Computer Applications

To

Guru Gobind Singh Indraprastha University, Delhi

Guide:
(Guide Name)
(Designation)

Submitted by:
(Student_name)
(Enroll No)

Institute of Information Technology & Management,
New Delhi – 110058
Batch (2023-2026)

Annexure-II

Certificate

This is to certify that this project entitled "……..." submitted in partial fulfillment of the degree of Bachelor of Computer Applications to the "……" through …… done by Mr./Ms…… ,Roll No…… is an authentic work carried out by him/her at under my guidance. The matter embodied in this project work has not been submitted earlier for award of any degree to the best of my knowledge and belief.

Signature of the student
Date:

Signature of the Guide:
Date:
Name of the Guide:
Designation:

Countersign HOD

Countersign Director

SELF CERTIFICATE

This is to certify that the dissertation/project report entitled "……" is done by me is an authentic work carried out for the partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications under the guidance of…….. The matter embodied in this project work has not been submitted earlier for award of any degree or diploma to the best of my knowledge and belief.

Signature of the student

Name of the Student

Roll No.

Annexure-III

TEMPLATE FOR CHAPTER SCHEME

C O N T E N T S

(Font size -18)

Certificate i
Acknowledgements ii
List of Tables/Figures/Symbols iii
Synopsis iv
CHAPTER 1: PROBLEM FORMULATION
1.1 Introduction about the Company 1
1.2 Introduction about the Problem 4
1.3 Present State of the Art 6
1.4 Need of Computerization 10
1.5 Proposed Software / Project 13
1.6 Importance of the work 15
CHAPTER 2 : SYSTEM ANALYSIS
2.1 Feasibility Study 20
2.1.1 Technical Feasibility 21
2.1.2 Economical Feasibility 22
2.1.3 Operational Feasibility 23
2.1.4 Other Feasibility Dimensions
2.2 Analysis Methodology 25
2.3 Choice of the Platforms 26
2.3.1 S/W used 28
2.3.2 H/W used

CHAPTER 3: SYSTEM DESIGN#
3.1 Design Methodology
3.2 UML Modeling 42
3.2.1 Use case
3.2.2 Sequence Diagram
3.2.3Activity Diagram
3.2.4 Class Diagram 32
3.3 Database Design 36
3.3.1 ERD 37
3.3.2 DFD 39
3.4 Input Design
3.5 Output Design 48
3.6 Code Design and Development

CHAPTER 4: TESTING AND IMPLEMENTATION##
4.1 Testing Methodology 52
4.1.1 Unit Testing 55
4.1.2 Module Testing 56
4.1.3 Integration Testing 57
4.1.4 System Testing 59

4.1.5 White Box / Black Box Testing 62
4.1.6 Acceptance Testing 69
4.2 Test Data & Test Cases 70
4.3 Test Reports and Debugging 71
4.4 Implementation Manual 73
4.5 Implementation 75
4.6 Users" Training 76
4.7 Post Implementation Maintenance 79

CHAPTER 5: CONCLUSION AND REFERENCES
5.1 Conclusion 81
5.2 System Specifications 83
5.2.1 H/W Requirement 83
5.2.2 S/W Requirement 84
5.3 Limitations of the System 85
5.4 Future Scope for Modification 86
5.5 References/Bibliography (as per format) 88

CHAPTER 6: ANNEXURES (Screen Snapshots must be annexed)
A-1 Menu Flow Diagram 92
A-2 Structure Chart 94
A-6 Decision Table/Tree 95
A-7 Data Dictionary 97
A-8 Test Reports 106
A-9 Sample Inputs 109
A-10 Sample Outputs 111
A-11 Coding (Optional) 118

Please note that for all the System Design (Database, Input & Output) the basic Prototype, format, Table Structure, etc. is to be discussed along with related validations, verifications & normalization. However, the sample Input & Output (Screen Snapshots) will be attached in the annexure.

Under Testing, you have to discuss the approach of Testing, Test Data, Test Cases and Test Report. How Debugging has been performed, on the basis of Test Report, must be also discussed?

Template for blank page before every chapter

CHAPTER 2

SYSTEM ANALYSIS

Feasibility Study

Technical Feasibility

Economical Feasibility

Operational Feasibility

Other Feasibility Dimensions

Analysis Methodology

Choice of the Platforms

S/W used

H/W used

Annexure-IV

LIST OF TABLES

Table No	Description	Page No
3.1	Employee's Personnel Information	18
3.2	Purchase Information	20
3.3	Stock Information	22
3.4	Master Table	26

LIST OF FIGURES

Figure No	Description	Page No
1.1	Organizational Chart	5
3.1	Data Entry Screen for Purchase	21
4.1		

LIST OF ABBREVIATIONS

Abbreviation	Description
CAD	Context Analysis Diagram
DFD	Data Flow Diagram
HIPO	Hierarchical Input Process Output

References/Bibliography:

1. Patterson D W, "Introduction to Artificial Intelligence and Expert Systems", Second Edition,2002, Prentice Hall of India Private Ltd., New Delhi.

2. V. Rajaraman, "An Introduction to Digital Computer Design", Third Edition, 1995, PrenticeHall of India Private Ltd., New Delhi.

Note: All of the above three will be prepared on separate pages.

Template for back-quote

Inventory Management System Student Name Enrollment Number

Annexure-IV

Guidelines for Data Analytics Using Python/ML/Deep learning
Major Project Topic Summary

Synopsis
1. Name / title of the project,
2. Statement about the problem
3. Significance of the Project (It must address Present State of the Art.)
4. Objective
5. Scope
6. H/W& S/W Specification
7. Data Collection and Methodology
8. Algorithm
9. Limitations / constraints of the project,
10. Conclusion, Future Scope for Modification,
11. References/Bibliography

CONTENTS

ABSTRACT
LIST OF FIGURES
LIST OF TABLES
LIST OF ABBREVATIONS
SYNOPSIS

Chapter -1 Introduction

1.1 Description of the topic: (Introduce the project's background and context)

1.2 Problem Statement: State the problem statement in a clear and concise manner. (**A problem statement is a short, clear explanation of an issue or challenge that sums up what you want to change**.)

1.3 Objectives: (Point wise objectives of the project)

1.4. Desired Outcomes (Point wise provide the desired outcomes in accordance to set objected)

1.5. Scope of the Project: (Cover the functionality and performance of the project, and targeted impact sector)

1.6. Project planning Activities

1.6.1 PERT Chart (A PERT chart is a tool used to schedule, organize, and map out tasks within a project.)

1.7. Organization of the report

Chapter 2 – Literature Review

Summary of Paper Studied (Study relevant theories, concepts, methodology used, and previous work related to your project and integrate a Literature review of at least 8 research papers of the related studies in a table, **Include only journal papers**.)

Briefly state the research gap in the existing literature.

Title of Research Paper	Author/Author(s) detail with year	Methodology Used	Findings

Chapter 3 – System Design and Methodology

3.1 System Design: (Explain the system architecture in detail using a schematic diagram)

- Briefly discuss the objectives of your research proposal.
- Briefly provide an overview of the proposed methodology.
- Use standard datasets only with at least 1000 data samples as per the application.
- Thoroughly provide a detailed description of the used dataset.

3.2 Algorithm Used (Describe in detail the algorithm you are using to address this problem and the motivation behind using it)

Chapter 4 – Implementation

2.1 Hardware and Software Requirement: (Explain the hardware and software Requirement used for your project)

4.2 Implementation Details: (Description of all modules in detail)

Chapter 5 –Results & Discussion

5.1. Results: (Use standard evaluation techniques to assess the performance of the proposed work).

5.2. Layouts: (Use visualizations, tables, and graphs to illustrate your findings effectively. The visualization graphs should be properly labeled and explicitly discussed in the text. Incorporate necessary layouts if applicable.)

5.3. Comparative Analysis: (Perform a detailed comparative analysis with the existing state-of-the art methods available in literature.)

Chapter 6 – Conclusion and Future Work

6.1 Conclusion: (Conclude the work using the following structure: background, aim, methodology, results and final concluding remarks.)

6.2. Future Scope: (Address the limitations of your project work. For each shortcoming, propose additions or enhancements that would help overcome it.)

References

Guidelines for IoT Major Project

Major Project Topic Summary

SYNOPSIS

1. **Title of Project**
2. **Problems with the Existing System: (Identify the shortcomings of the manual system and how the proposed system addressed them)**
   (Note: Consider the previously existing project as Manual)
3. **Related Work: (Review at most recent ten research papers related to the field of study –consider last 3 years papers)**
4. **Description of the Proposed System**
5. **Tools/Platform/Middleware**
6. **Future Scope**

CONTENT OF THE FILE (Chapter wise)

CHAPTER 1- INTRODUCTION

1.1 Overview of the Project
1.2 Motivation of the Project
1.3 Research gap
1.4 Research Objective
1.5 Structure of the report
1.6 Tools/Platforms
1.6.1 Hardware Specification tools
1.6.2 Software Specification Tools
1.6.3 Any additional hardware and software requirement along with the versions can be included. (Optional)
1.7 Description of Proposed System
1.8 Project planning Activities
1.5.1 Gantt Chart
1.5.2 Risk Management and Mitigation Strategies

CHAPTER 2- LITERATURE REVIEW

2.1 Summary of Paper Studied: (Precisely summarize each research paper reviewed, highlighting their contributions to the project domain.)
2.2 Integrated summary of the Literature studied.(Comparative analysis in Tabular form)

Title of Research Paper	Author/Author(s) detail with year	Methodology Used	Findings

CHAPTER 3- EXPERIMENTAL SETUP

3.1 System Overview
3.2 Block Diagram
3.3 System Hardware (Description of the hardware)
- **3.3.1 Raspberry Pi/Arduino UNO (General information with diagram)**
- **3.3.2 Sensors and Actuators (General information with diagram)**
- **3.3.3 Proposed System Module Description**
- **3.3.4 Circuit Diagram(Images to be attached)**
- **3.3.5 Pin Diagram (Images to be attached)**
- **3.3.6 Hardware Description (In detail: Include Features, Specifications)**

3.4 System Software (Description of the software)
- **3.4.1 Description of System Software: (Discuss software components, middleware, protocols, and APIs utilized.)**
- **3.4.2 Flow Chart**
- **3.4.3 IoT: IDE description (respective to proposed system example: Arduino uno, Raspberry pi)**
- **3.4.4 Steps involved in: Title of Project (i.e. working of Proposed System)**
- **3.4.5 Program Code**

CHAPTER 4- RESULT ANALYSIS

4.1 Results: Snapshots: (Include snapshots and provide qualitative and quantitative analysis of the system's performance.) input and output modules.)

CHAPTER 5- CHALLENGES.FUTURE SCOPE AND CONCLUSION

5.1 Challenges (Discuss technical, logistical, and implementation challenges encountered during the project.)

5.2 Limitations

5.3 Applications (Explore potential real-world applications of the proposed system beyond the immediate scope.)

5.4 Future Scope (Discuss potential enhancements, scalability, and adaptability of the proposed system.)

5.5 Conclusion (Summarize key findings, contributions, and implications of the project.)

References (IEEE FORMAT)

Appendixes